DIGITAL MARKETING ANALYTICS

Web Analytics Dashboard and Performance Report

Google Analytics 4: Google Merchandise Store Demo Account

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Executive summary

This project looks at how the Google Merchandise Store performed in Google Analytics 4 (GA4) over a window that the GA4 charts show as roughly August to October 2026. Using the demo account data, we built an Excel dashboard and read six GA4 reports (audience, acquisition, engagement, conversion, pages and campaigns) to find out where visitors come from, how they behave and where the money is made.

In total the store recorded **292,824 users, 388,430 sessions, 5.06 million events and $812,240 in revenue**, with an overall engagement rate of 41.76%. Five findings stand out:

- **Traffic is heavily concentrated.** Direct accounts for 57.9% of sessions and 51.6% of revenue, but only 28.5% of its sessions are engaged. Direct plus Unassigned together make up 63% of all sessions, so source tracking needs attention.
- **Organic Search is the strongest large channel.** It delivers 25.0% of sessions and 34.6% of revenue with a 70.6% engagement rate.
- **Small channels are the most valuable per visit.** Organic Social ($7.21), Referral ($6.23) and Organic Video ($5.97) earn roughly three times the store average of $2.09 revenue per session, yet together they are about 3% of traffic.
- **Paid Search underperforms.** It supplies 8.3% of sessions but 2.9% of revenue ($0.74 per session) and holds visitors for only 17 seconds on average.
- **The home page loses people quickly.** It takes 25% of all page views but averages 9 seconds of engagement and has a 63.1% bounce rate, while category pages bounce at only 9% to 15%.

Several numbers need to be verified before they are used for decisions: a sharp spike in product-list events around 28 September 2026, checkout-page revenue that equals 97% of total revenue, and campaign cost data that shows zero. We treat them as measurement questions, not as results. The recommendations in Section 6 start with fixing these and then move to organic search, paid search quality, partnerships, home-page navigation and email.

Introduction and objectives

Every online retailer collects a large amount of behavioural data, but the data only becomes useful once someone turns it into a few clear questions and answers. This project does that for a real, publicly available e-commerce dataset: the Google Merchandise Store demo account in Google Analytics 4. The store sells branded apparel, drinkware and accessories, which makes it a good practice case for acquisition, engagement and conversion analysis.

The project has the following objectives:

- Use the Google Analytics demo account dataset as the data source.
- Generate and interpret at least five analytics reports covering audience, acquisition, engagement, conversion and pages/campaigns.
- Build a dashboard in Excel with KPI cards, charts and performance metrics.
- Explain what the findings mean for managers.
- Recommend strategic actions that are backed by the numbers.

Data source and approach

The data comes from the public GA4 demo account for the Google Merchandise Store. We took the values shown in the GA4 User acquisition, Traffic acquisition, Events, Pages and screens, Non-Google campaign and Reports snapshot views and entered them into an Excel workbook, one sheet per report. We did not change any GA4 number. Where two GA4 reports show slightly different totals for the same measure (for example 5,058,914 events in Traffic acquisition against 5,058,644 in User acquisition), both are kept as displayed and the difference is noted.

Everything on the dashboard is formula-linked to those sheets. A separate Chart Data sheet adds four derived measures that GA4 does not show directly: share of sessions, share of revenue, revenue per session (revenue divided by sessions) and a revenue index (share of revenue divided by share of sessions; above 1.0 means a channel earns more than its traffic share). Table 1 lists the reports used.

GA4 report	Worksheet	Used for
User acquisition	Audience Analysis	Users, new vs returning, engagement time per user
Traffic acquisition	Channel Performance	Sessions, engagement rate, key events, revenue by channel
Events	Events & Conversions	Event volumes, key events by source, purchases
Pages and screens	Pages & Screens	Views, engagement time and revenue by page path
Non-Google campaign	Campaigns	Campaign-level sessions, ads clicks, cost, ROAS
Reports snapshot	Reports Snapshot	Top sources, cities, audiences, bounce rates

Table 1: GA4 reports used and where they sit in the workbook

Dashboard overview

The dashboard (Figures 1 and 2) opens with eight KPI cards and then shows nine charts. All values below are the GA4 summary totals.

Total users	Sessions	Engagement rate	Total revenue
292,824	**388,430**	**41.76%**	**$812,240.44**
Event count	**Key events**	**User key-event rate**	**Avg engagement / user**
5,058,914	**374,311**	**21.12%**	**1m 03s**

Table 2: Headline KPIs from the dashboard

Two ratios give useful context for these totals. The average session brings in about **$2.09** of revenue ($812,240 divided by 388,430 sessions), and 92% of users (269,888 of 292,824) are new. In other words the store is very good at attracting first-time visitors and has much less evidence of bringing them back.

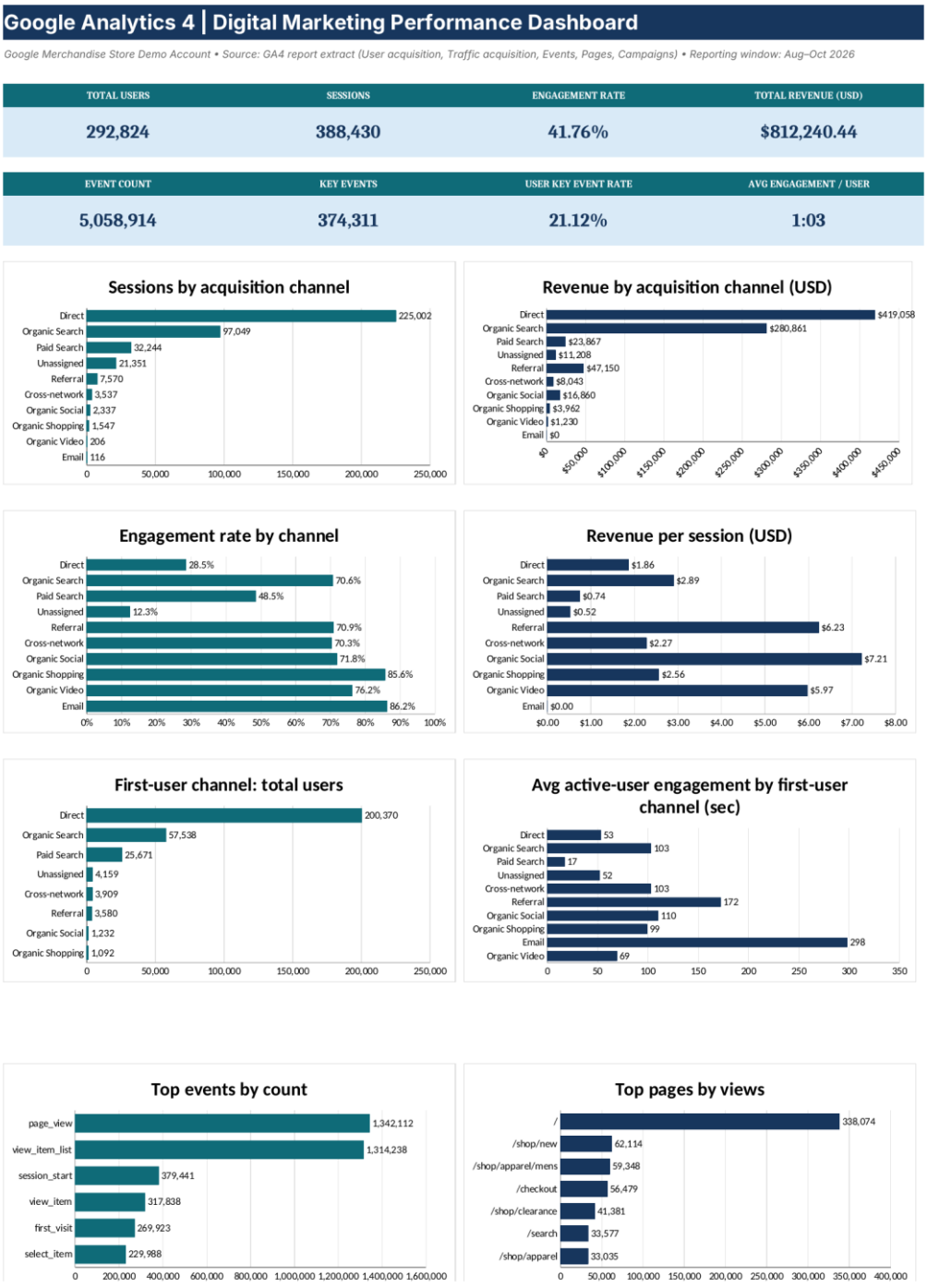


Figure 1: Excel dashboard, KPI cards and channel, audience, event and page charts (screenshot of the workbook)

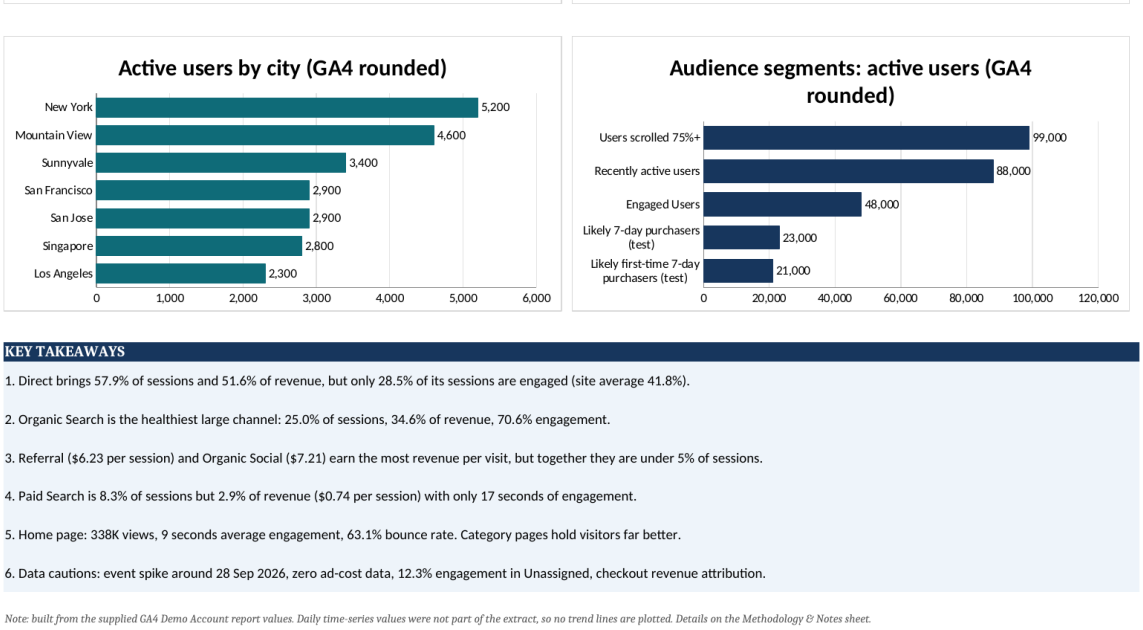


Figure 2: Excel dashboard, audience segments, cities and key takeaways (screenshot of the workbook)

Analytics reports and interpretation

Each report below follows the same pattern: what we looked at, what the numbers say, and what we make of them.

Audience report

What we looked at. Total users, new versus returning users, engagement time per user and the most active locations and audience segments.

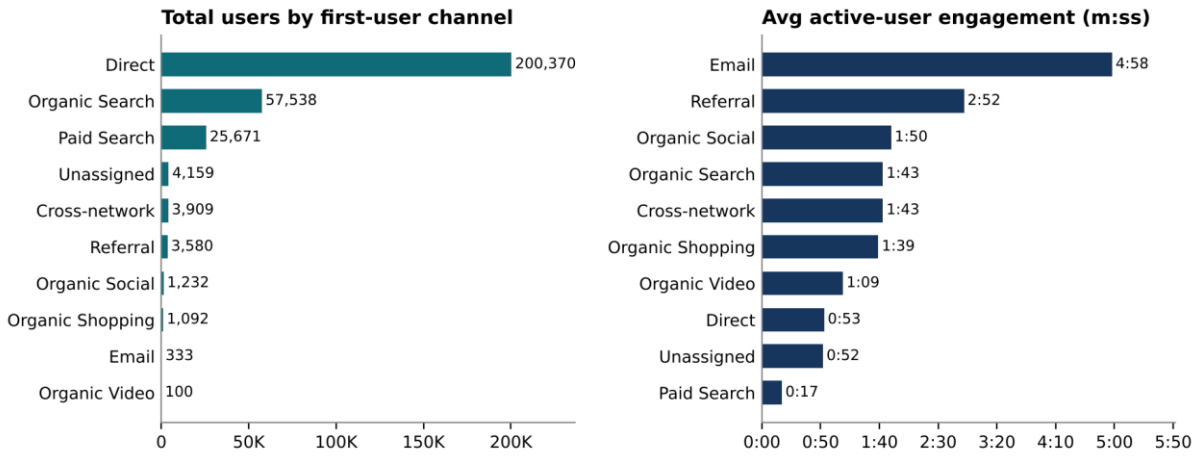


Figure 3: Users and engagement time by first-user channel (GA4 User acquisition)

What the data shows. The store had 292,824 users, of whom 269,888 (92.2%) were new. Direct was the first-touch channel for 200,370 users (68.4% of the total), followed by Organic Search with 57,538 (19.6%) and Paid Search with 25,671 (8.8%). Engagement time varies a great deal by where users first arrived: Email users stayed 4m 58s on average, Referral users 2m 52s and Organic Search users 1m 43s, against 53 seconds for Direct and only 17 seconds for Paid Search. Email also had the highest user key-event rate (65.8%) and 308 of its 333 users were returning visitors.

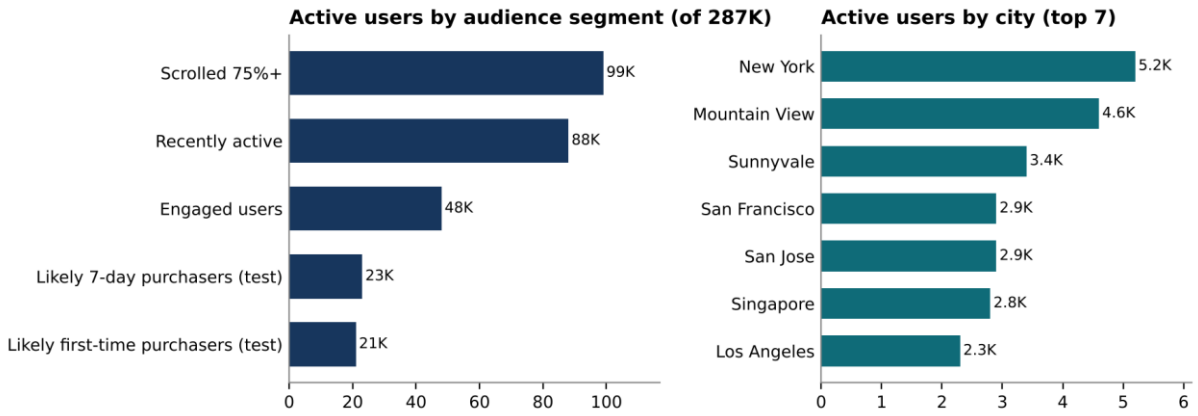


Figure 4: Active users by GA4 audience segment and by city (rounded values from the Reports snapshot)

The snapshot lists 99K users who scrolled at least 75% of a page (about 34% of the 287K active users), 48K engaged users and 23K "likely 7-day purchasers", a predictive audience that GA4 marks as a test. Geographically, the top seven cities add up to only about 24K users (roughly 8% of active users), led by New York, Mountain View and Sunnyvale, so the audience is widely spread rather than clustered in a few places. The report notes that city and user totals are not strictly additive, so these shares are approximate.

> **Interpretation**
> The store has a large audience of one-time visitors and a very small group of loyal, high-value ones. The users who come through Email and Referral are the most attentive, but there are few of them. Growing the number of people who come back, not just the number who arrive, is the clearest audience opportunity.

Acquisition report

What we looked at. Sessions, revenue and revenue per session for each default channel group in GA4 Traffic acquisition.

Channel	Sessions	% of sessions	Revenue (USD)	% of revenue	Revenue / session
Direct	225,002	57.9%	419,058	51.6%	$1.86
Organic Search	97,049	25.0%	280,861	34.6%	$2.89
Paid Search	32,244	8.3%	23,867	2.9%	$0.74
Unassigned	21,351	5.5%	11,208	1.4%	$0.52
Referral	7,570	1.9%	47,150	5.8%	$6.23
Cross-network	3,537	0.9%	8,043	1.0%	$2.27
Organic Social	2,337	0.6%	16,860	2.1%	$7.21
Organic Shopping	1,547	0.4%	3,962	0.5%	$2.56
Organic Video	206	0.1%	1,230	0.2%	$5.97
Email	116	0.0%	0	0.0%	$0.00
All channels	**388,430**	**100%**	**812,240**	**100%**	**$2.09**

Table 3: Channel performance (GA4 Traffic acquisition)

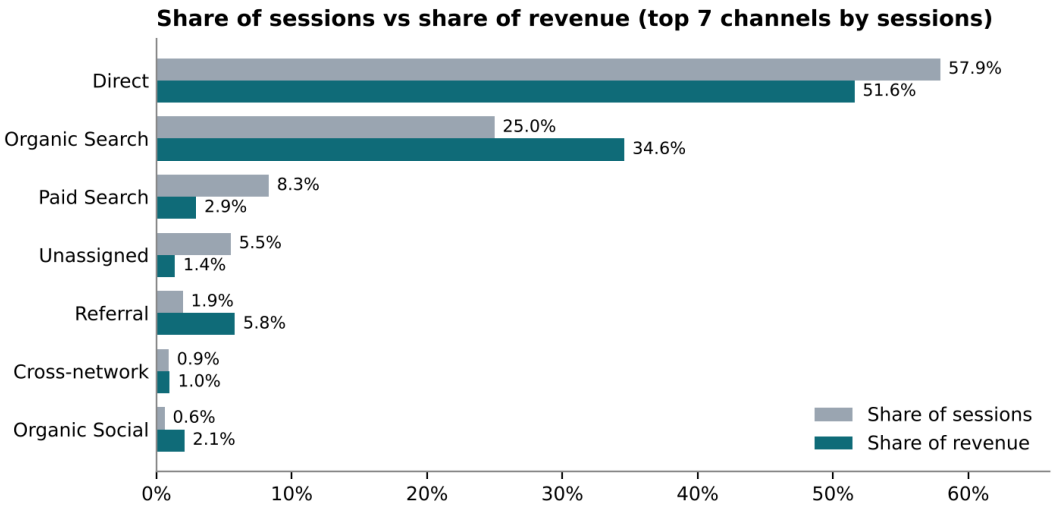


Figure 5: Share of sessions compared with share of revenue

What the data shows. Direct and Organic Search together bring in 83% of sessions and 86% of revenue. Direct is the largest by volume, but its revenue share (51.6%) is smaller than its session share (57.9%), so each Direct session is worth less than the store average. Organic Search does the opposite, earning 34.6% of revenue from 25.0% of sessions (a revenue index of 1.38). Paid Search is the weakest of the major channels: 8.3% of sessions, 2.9% of revenue, and a revenue index of just 0.35. Referral (index 2.98), Organic Social (3.45) and Organic Video (2.86) are far above 1.0 but small in volume.

One more thing stands out. Direct and Unassigned together make up 246,353 sessions, which is 63.4% of all traffic. "Direct" in GA4 often includes visits that arrived without a source being recorded (untagged links, some apps, bookmarks), and "Unassigned" means GA4 could not classify the visit at all. We cannot say with confidence where most of the store's traffic really comes from.

Interpretation

Revenue is concentrated in Direct and Organic Search, which makes the store dependent on brand awareness and search visibility. The channels that earn the most per visit (Referral, Social, Video) are under-used. Paid Search is spending effort, and probably money, on traffic that does not convert at the rate of the rest of the site.

Engagement report

What we looked at. Engagement rate (engaged sessions divided by sessions), average engagement time, events per session and page-level engagement.

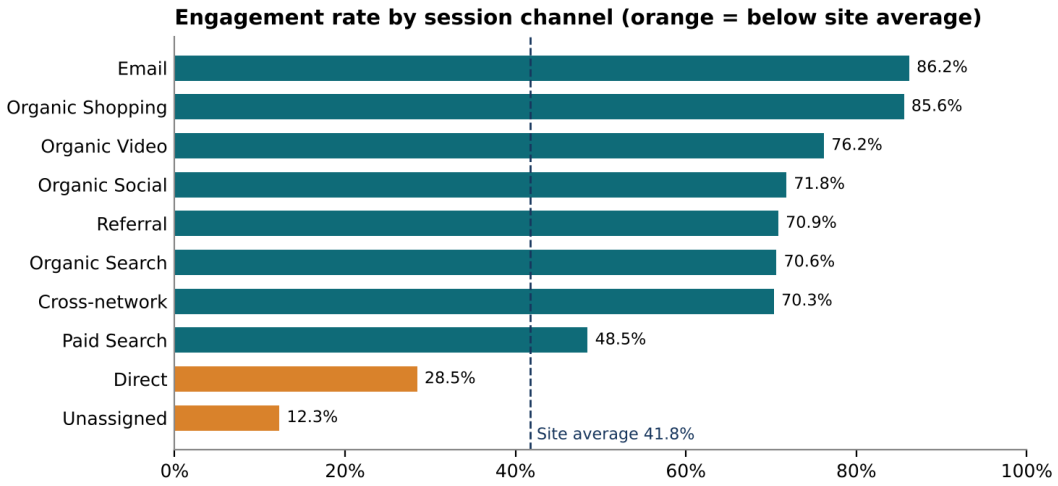


Figure 6: Engagement rate by channel, compared with the 41.76% site average

What the data shows. The overall engagement rate is 41.76% (162,203 engaged sessions out of 388,430). Direct (28.5%) and Unassigned (12.3%) are the two channels clearly below that line, and both are large, which is what drags the site average down. Every organic and referral channel sits between 70% and 86%. Paid Search is close to the average at 48.5%, but its average session engagement time is the shortest of any channel at 17 seconds, and it has the fewest events per session (6.91 against a site average of 13.02).

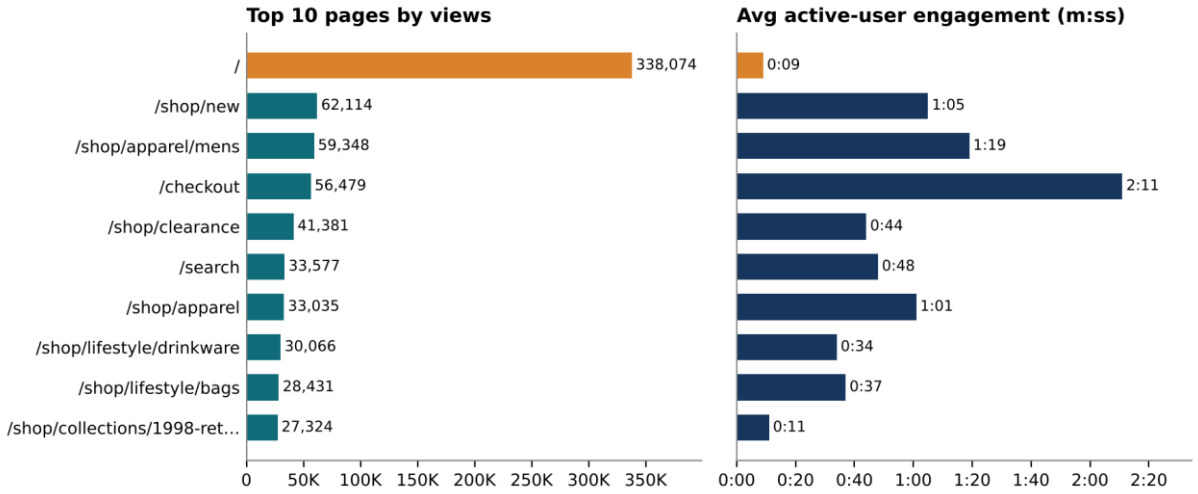


Figure 7: Top 10 pages by views and by average active-user engagement time

At page level the home page (/) is the single biggest source of views (338,074, or 25.2% of the 1.34 million total) but people spend only 9 seconds there on average, and the snapshot gives it a 63.1% bounce rate. By contrast, the New (/shop/new, 1m 05s), Men's apparel (1m 19s) and checkout (2m 11s) pages hold attention for much longer, and the category pages in the snapshot show bounce rates of just 9.0% (Sale) to 15.3% (New). The retro collection page is another weak spot at 11 seconds.

Interpretation

Visitors who reach a product category tend to stay and browse. The problem is the step before that: a large share of the audience lands on the home page, glances at it and leaves. A short time on the home page is not bad by itself, because people may click straight through, but combined with the high bounce rate it suggests the home page is not guiding people well enough into the shop.

Conversion and events report

What we looked at. Key events, purchases, revenue, event volumes and the event spike flagged by GA4.

What the data shows. GA4 recorded 374,311 key events, a user key-event rate of 21.12% and a session key-event rate of 23.2%. These rates look high, and it is important not to read them as sales conversion. In this account the key events include view_item and add_to_cart as well as purchase, so they measure meaningful interaction, not completed orders. The purchase event is the one to watch: the extract lists 2,695 purchases from the direct source, generating $419,058 (an average order value of about $155.49). If we treat that as a lower bound, at least 0.7% of sessions ended in a purchase (2,695 of 388,430). Among Direct visitors, 27,326 add-to-cart events against 2,695 purchases means roughly one in ten carts led to an order (9.9%).

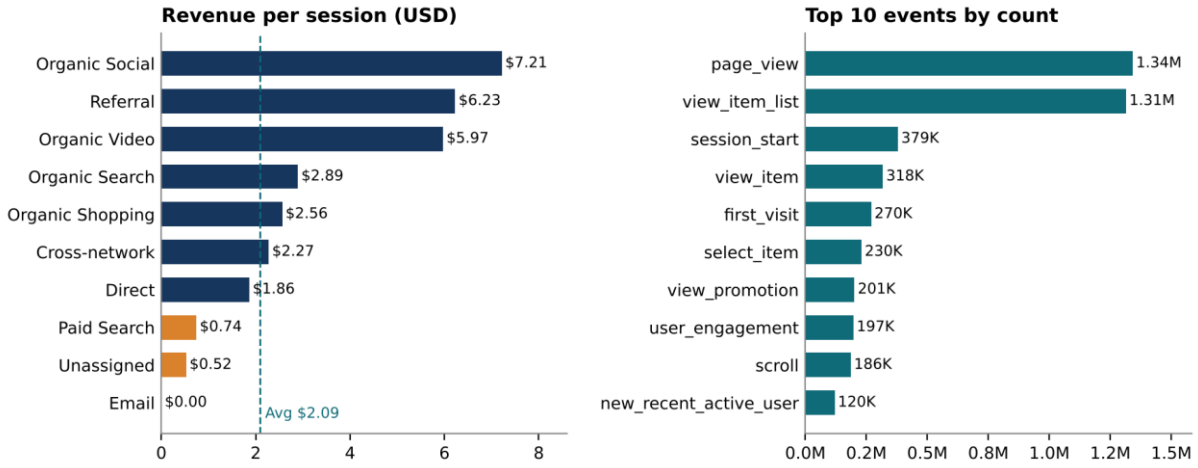


Figure 8: Revenue per session by channel and the ten most frequent events

The event list is dominated by page_view (1.34 million) and view_item_list (1.31 million). That second number is unusual: it works out at 15.2 list views per user, and GA4's own automated insight flags a sharp jump around 28 September 2026, with weekly view_item_list events going from about 15K to 125K and view_item key events from about 3.1K to 35K. A jump of that size can come from a genuine campaign or promotion, but it can just as easily come from a tagging change, duplicate events or bot traffic.

There is a second data question on the checkout page. The /checkout path shows $791,527 of revenue, which is 97.5% of the $812,240 total. That tells us purchase revenue is being attributed to the page where the purchase event fires, not that checkout generates sales in any independent sense.

> **Interpretation**
> The store makes money, and revenue per session ranges from $0.52 to $7.21 depending on channel. But the conversion picture is clouded by three measurement issues (key-event definitions, the 28 September spike and checkout revenue attribution). Until these are checked, conversion rates should be quoted with care.

Campaign report

What we looked at. Session campaign, ads clicks, ad cost and return on ad spend (ROAS).

What the data shows. The GA4 campaign report carries an "insufficient data" notice. Ads clicks, cost and impressions are all zero, which makes ROAS 0.00 for every row. Sessions in the table total 110,000, only about 28% of the 388,430 sessions in the channel report. The two default rows, (organic) at 98,104

sessions and (referral) at 10,601, make up almost all of that. The named campaigns are tiny: the largest, Aug025_Noogler_V1, has 98 sessions, of which 84 (86%) were engaged, and the others have fewer than ten sessions each.

Interpretation

No conclusion can be drawn about paid-media return from this table. A ROAS of zero here reflects missing cost data (most likely Google Ads is not linked or the demo data does not include it), not proof that campaigns earn nothing. It also shows that campaign tagging is very thin: almost no traffic carries a named campaign, which is also part of why Direct and Unassigned are so large.

Managerial implications

- **Dependence on two channels is a business risk.** Direct and Organic Search generate 86% of revenue. A drop in brand demand or a search-algorithm change would hit revenue hard, so diversification matters.
- **Decisions about budget should wait for cleaner attribution.** With 63% of sessions sitting in Direct or Unassigned, managers cannot see which marketing activity is actually working. Better UTM tagging is a low-cost step that improves every later decision.
- **Paid Search needs a quality review before more budget.** Its revenue per session is about a third of the store average. Spending more on the same set-up would likely scale up the same weak result.
- **High-value, low-volume channels deserve experimentation.** Referral and social visitors are worth roughly three times an average visitor. Because the base is small, the right move is a controlled test, not a large commitment.
- **The home page is the main bottleneck in the browsing journey.** It is the most-viewed page, and improving the click-through from it to category pages touches a quarter of all page views.
- **Retention is under-developed.** With 92% new users and a tiny but very engaged email audience, there is room to invest in CRM and remarketing, provided consent rules are followed.
- **Trust in the numbers comes first.** The event spike, checkout revenue attribution and consent-mode warning (GA4 reports "lost behavioural metrics") mean the analytics set-up itself needs an audit.

Strategic recommendations

Table 4 sets out the recommendations in the order we would tackle them. Each one points back to evidence in the data, and the right-hand column names the GA4 measure to watch. We use current values as baselines rather than inventing targets, because the demo account has no business goals attached.

#	Recommendation	Evidence from the data	KPI to track (baseline)	Priority
1	Fix measurement first: audit UTM tags and channel rules, check the 28 Sep event spike, validate purchase and revenue events, review consent mode.	Direct + Unassigned = 63.4% of sessions; Unassigned engagement 12.3%; checkout page shows 97.5% of revenue; ads cost is zero.	Share of sessions in Unassigned (5.5%); consistent event totals across reports	High, 0 to 30 days
2	Protect and expand Organic Search: strengthen category and product content, internal links and landing pages that already rank.	25.0% of sessions, 34.6% of revenue, 70.6% engagement, revenue index 1.38.	Organic sessions (97,049); organic revenue ($280,861)	High, 30 to 90 days
3	Fix Paid Search quality before raising budget: review search terms, add negative keywords, send traffic to category pages instead of the home page, and link Google Ads cost data.	8.3% of sessions, 2.9% of revenue, $0.74 per session, 17 sec engagement, fewest events per session.	Revenue per session ($0.74); engagement time (0:17)	High, 0 to 60 days
4	Run partner, referral and social tests: identify the top referring sites and test creator, partner and social placements.	Referral $6.23 and Social $7.21 per session against $2.09 average; both under 3% of traffic.	Referral + social sessions (9,907); revenue per session	Medium, 30 to 90 days
5	Redesign home-page navigation: A/B test hero banners and links to New, Men's and Sale, and add clearer category entry points.	Home: 25.2% of views, 9 sec engagement, 63.1% bounce; category pages 9% to 15% bounce.	Home bounce rate (63.1%); click-through to category pages	Medium to high, 30 to 60 days
6	Diagnose the cart-to-purchase drop: build a GA4 funnel exploration (view item, add to cart, begin checkout, purchase) and test checkout friction.	Direct: 27,326 add-to-cart vs 2,695 purchases (9.9%); checkout page engagement 2:11.	Cart-to-purchase rate (9.9% for Direct)	Medium to high, 30 to 90 days
7	Grow the email and returning-customer base with consent-based sign-up prompts and post-purchase flows.	Email: 4:58 engagement, 65.8% user key-event rate, 308 of 333 users returning; overall 92% new users.	Returning users (42,093); email users (333)	Medium, 60 to 120 days
8	Use GA4 audiences for remarketing (with consent): scrolled 75%+, engaged users and likely purchasers.	99K scrolled 75%+; 48K engaged; 23K likely 7-day purchasers (test).	Audience size; remarketing revenue per session	Medium, after Rec. 1

Table 4: Recommendations, supporting evidence and KPIs

Recommendations 1 and 3 should start immediately because they cost little and protect the quality of every other decision. Recommendations 2 and 5 are where we would expect the most revenue upside from the data we have, since they affect the largest and most-viewed parts of the site. All of them are hypotheses to test, not guaranteed outcomes, and each should be measured against the baseline in the table before it is judged a success.

Limitations

- **Reporting window.** The GA4 charts show August to October 2026, but the exact start and end dates were not captured in the extract. They should be confirmed in GA4 before the work is cited as time-specific.
- **No daily trend data.** The extract has totals and breakdowns but no day-by-day values, so the dashboard has no trend lines and we have not estimated any. The 28 September spike is reported only as GA4 described it.
- **Demo data.** The Google Merchandise Store is a demonstration account. Its patterns (for example, a very large Direct share) may not reflect a typical retailer, and some behaviour may be the result of how the demo data is generated.

- **Rounded and non-additive figures.** Snapshot values such as 287K, cities and audiences are rounded by GA4, and user counts by channel do not add up exactly to the total because of how GA4 attributes users.
- **Purchases are only partly visible.** The key-event table lists one purchase row (Direct), so order counts and average order value are lower-bound estimates for that source, not store-wide figures.
- **Campaign and cost data are missing.** ROAS and cost per key event could not be calculated.
- **Consent mode.** GA4 shows a recommendation to reconfigure consent mode to recover lost behavioural metrics, so some behaviour may be unmeasured.

Conclusion

The GA4 data tells a fairly clear story. The Google Merchandise Store attracts a lot of first-time visitors, earns most of its revenue from Direct and Organic Search, and gets its best value per visit from a handful of small channels that it barely uses. Paid Search and the home page are the two places where performance is clearly below what the rest of the site shows. At the same time, a lot of the traffic cannot be traced to a source, and a few key numbers (the September spike, checkout revenue and ad cost) need checking before anyone acts on them.

For a manager the order of work is straightforward: clean up the measurement, protect organic search, repair paid search, test partner and social channels, simplify the path from the home page into the shop, and look at the cart-to-purchase drop. The Excel dashboard that accompanies this report keeps all of the numbers linked to their sources, so the analysis can be refreshed when new GA4 data is available.